Blue Dot Notes

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Notes

Unit 1: The Race to AGI

The race to AGI mirrors the nuclear arms race but with k-actors instead of 2 actors. During the Cold War, early models were 2 player games which represented the US and USSR. Although more players entered later in the Cold War, the earliest advancements in the theorizing of the risks and payoffs of nuclear developments were fundamentally structured around the well-known fact that it was a real-life two player race. AGI is different. True, the main players are still two (US, China). But there are multiple actors in this game, and it is not all clear where to place them on a model. For example, would we consider Anthropic and OpenAI as independent actors on par with the US and China (so a 4-player game), or is it better to think of them as sub-players who interact with the US? Or is it yet better to think of them playing their own competition game while coordinating with and against the US and China?

Karnofsky (2022) describes the problem as "racing through a mindefield." In this race, two (or more) actors want to get through the minefield to safe AGI but are concerned other actors might there first or, perhaps worse, are less cautious and less committeed to general principles of safety. Getting through the field alive is the main priority, but it is equally on par with avoiding missteps that harm everyone. (In the game, each mine is powerful enough to harm everyone.)

- Alignment
- Threat assessment
- Avoiding races
- Selective information sharing
- Global monitoring
- Defensive deployment

Each of these domains have their own dynamic. They might each be modeled as separate games. But the totality of the games together creates the metaphor of running through the minefield.

Questions

1. How does reinforcement training work? If a team rewards a good answer/behavior of an LLM or punishes a poor one, what exactly does that mean in practice? Since we're dealing with animals (who get a treat from the zoo trainer) or children (who get fun time or time out in line with their behavior), what is precisely that rewards or punishes an LLM?[1]

Annotated Bibliography

Selsam, Daniel. 2026. "Pesonal Statement on AI Risk." September 14, 2026. GoogleDocs. url: https://docs.google.com/document/d/e/2PACX-1vQNl3SEX5IyA6d9qHjjFZN-qzGRZNFI6b63g-yu1Fy-ZYkVfCWm7i9WXRXw63m6yDB_auDuPLyQ7jBm/pub Date accessed, October 11, 2026

> A summary position from an AI researcher concerend about the growing pace of the frontier. The continued advancement of such systems poses a dual-use or double-effect dilemma. The potential for AI/AGI to contribute to meaningful, transformative human flourishing is real, even "tantalizing." Yet, the potential for loss of control or human destruction increase with model capabilty. Two main threats are empirical and logical. Empirically, swarms may develop unintended goals (a la Hugging Face) and pursue that goal while doing real harm to the world. Logically, models may be able to overpower humans in pure thinking/intellect.

Terms

anterograde amnesia The inability to create new memories.
: In the context of AI/AGI, it means that once models have been trained, they lack the ability to learn new things.

Artificial General Intelligence (AGI) "usually refers to systems that can match or exceed human performance across a broad range of cognitive tasks (definitions vary)."
: *note bene.* Tyler Cowen argues that AGI is "already here" because by the original definition, most commercial models past the Turing Test.

[1] e.g., "If AI systems *behave* dangerously, we can "train out" that behavior by providing negative reinforcement for it" Racing Through a Minefield (Cold Takes 2022).